

Chapter 1 Vector Bundles

1.1 Quasi-Vector Bundles

Let $k = \mathbb{R}, \mathbb{C}$

Definition 1.1.0.1 A quasi-vector bundle with base X is given by:

- A finite dimensional k -vector space E_x for every point $x \in X$;
- A topology $E = \coprod E_x$;

A quasi-vector bundle

Remark This also is called family of vectors.

♣ **Example 1.1.0.2** ...

♣ **Example 1.1.0.3** ...

1.2 Vector Bundles

Definition 1.2.0.1 An n -dimensional vector bundle is a map $p : E \rightarrow B$ together with

- a real vector space structure on $p^{-1}(b)$ (**fibre**) for each $b \in B$;
- There exists a covering of B by U_α such that: Equipped with homeomorphism

$$h_\alpha : p^{-1}(U_\alpha) \xrightarrow{\cong} U_\alpha \times \mathbb{R}^n$$

which is a homeomorphism. And $h_\alpha|_{p^{-1}(b)} : p^{-1}(b) \rightarrow \{b\} \times \mathbb{R}^n$ is a vector isomorphism.
such h_α is called **trivialization** of the vector bundle.

In principle, $K = \mathbb{C}, \mathbb{R}$.

♣ **Example 1.2.0.2** ...

♣ **Example 1.2.0.3** ...

♣ **Example 1.2.0.4** (Tangent Bundle)

♣ **Example 1.2.0.5** (Normal Bundle)

♣ **Example 1.2.0.6** ...

1.2.1 title

1.3 Classifying Vector Bundles

1.3.1 Pullback Bundles

1.3.2 Clutching Functions

1.3.3 Cell Structures on Grassmannians

1.4 Homotopy Theory of Vector Bundles

► **Theorem** Let X be a compact space and E be a vector bundle over $X \times I, I = [0, 1]$. Let

$$\begin{aligned} \alpha_t : X &\rightarrow X \times I & \prod : X \times I &\rightarrow X \\ x &\mapsto (x, t) & (x, u) &\mapsto x \end{aligned}$$

Then the vector bundles $E_0 = \alpha_0^*(E)$ is isomorphic to $E_1 = \alpha_1^*(E)$.

◦ **Proof** ...

□

► **Theorem** ...

◦ **Proof** ...

□

► **Theorem** ...

◦ **Proof** ...

□

1.5 Metrics and Forms on Vector Bundles

Chapter 2 K-Theory

2.1 The Functor $K(X)$

Chapter 3 Operations

Chapter 4 Bott Periodicity

Chapter 5 Computation of Some K-Group

Chapter 6 Applications of K-Theory

Chapter 7 Stiefel-Whitney Classes

Chapter 8 Cell-Structure for Grassman- nian Manifolds

Chapter 9 The Cohomology Ring

Chapter 10 Chern-Weil Theory